

Dataset Overview

Dataset Description: The IBM HR Analytics Employee Attrition & Performance dataset contains 1,470 records and 35 evaluation columns tracking demographic, work-life, and financial variables.

Business Domain: Human Resources & Talent Management analytics.

Business Problem: Unplanned employee attrition creates replacement costs, destabilizes projects, and lowers team morale.

Target Variable: Attrition (Binary: Yes / No).

Project Objectives: Identify key attrition drivers, build a machine learning pipeline to predict employee departure risk, and deliver actionable operational strategies.

. Data Preparation

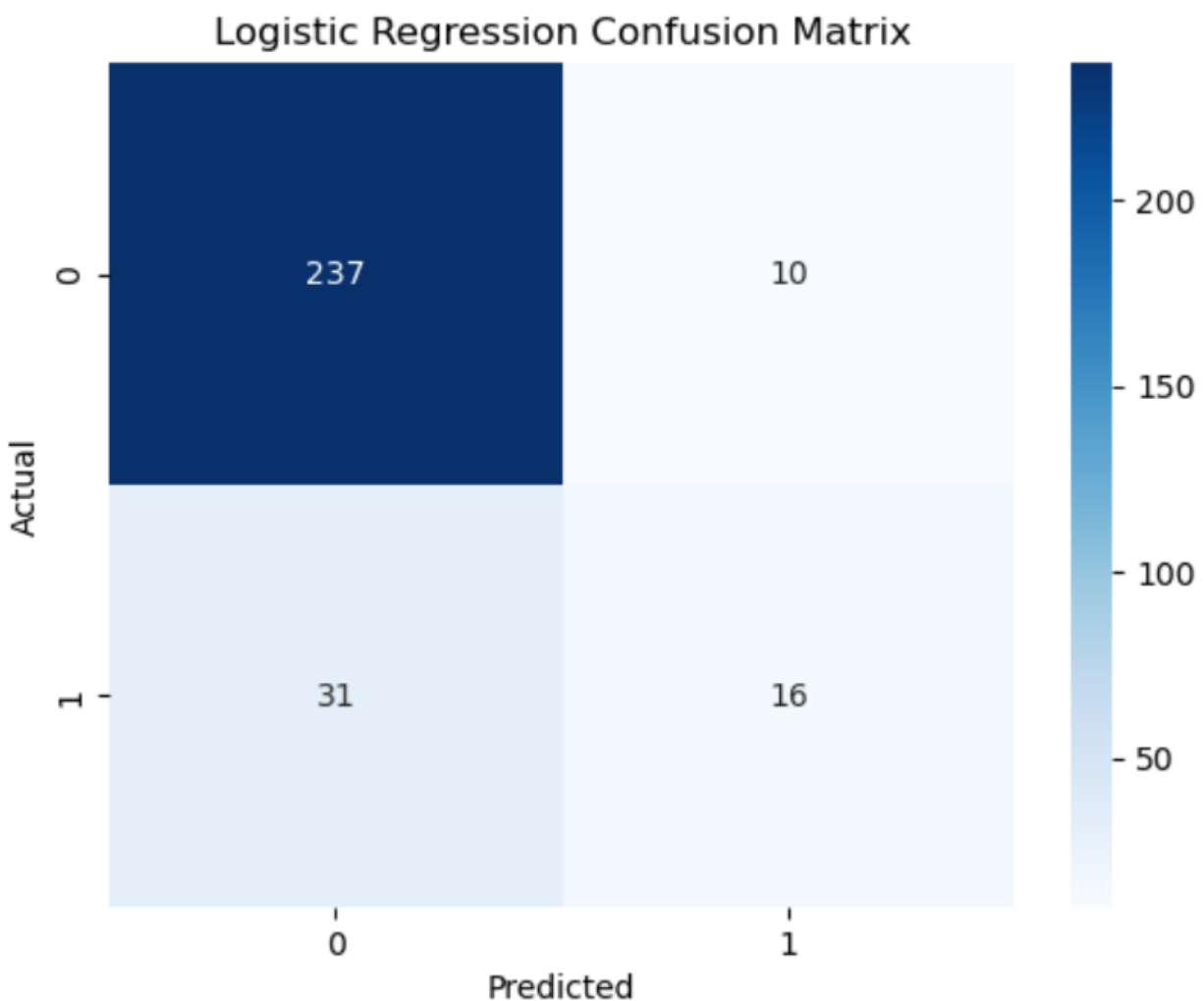
- **Data Quality Issues:** Zero missing values detected across 1,470 records. Constant/zero-variance columns (EmployeeCount, StandardHours, Over18) added no information value.
- **Cleaning Process:**
 - Removed uninformative features (EmployeeCount, StandardHours, Over18, EmployeeNumber).
 - Converted Attrition into binary values (Yes = 1, No = 0).
- **Feature Engineering:** Standardized continuous features (Age, MonthlyIncome, DistanceFromHome) using StandardScaler to ensure normalized convergence for gradient models. Categorical variables were encoded using LabelEncoder.
- **AI-Assisted Preprocessing:** Automated column profiling removed zero-variance attributes and detected feature correlation flags across demographic dimensions.

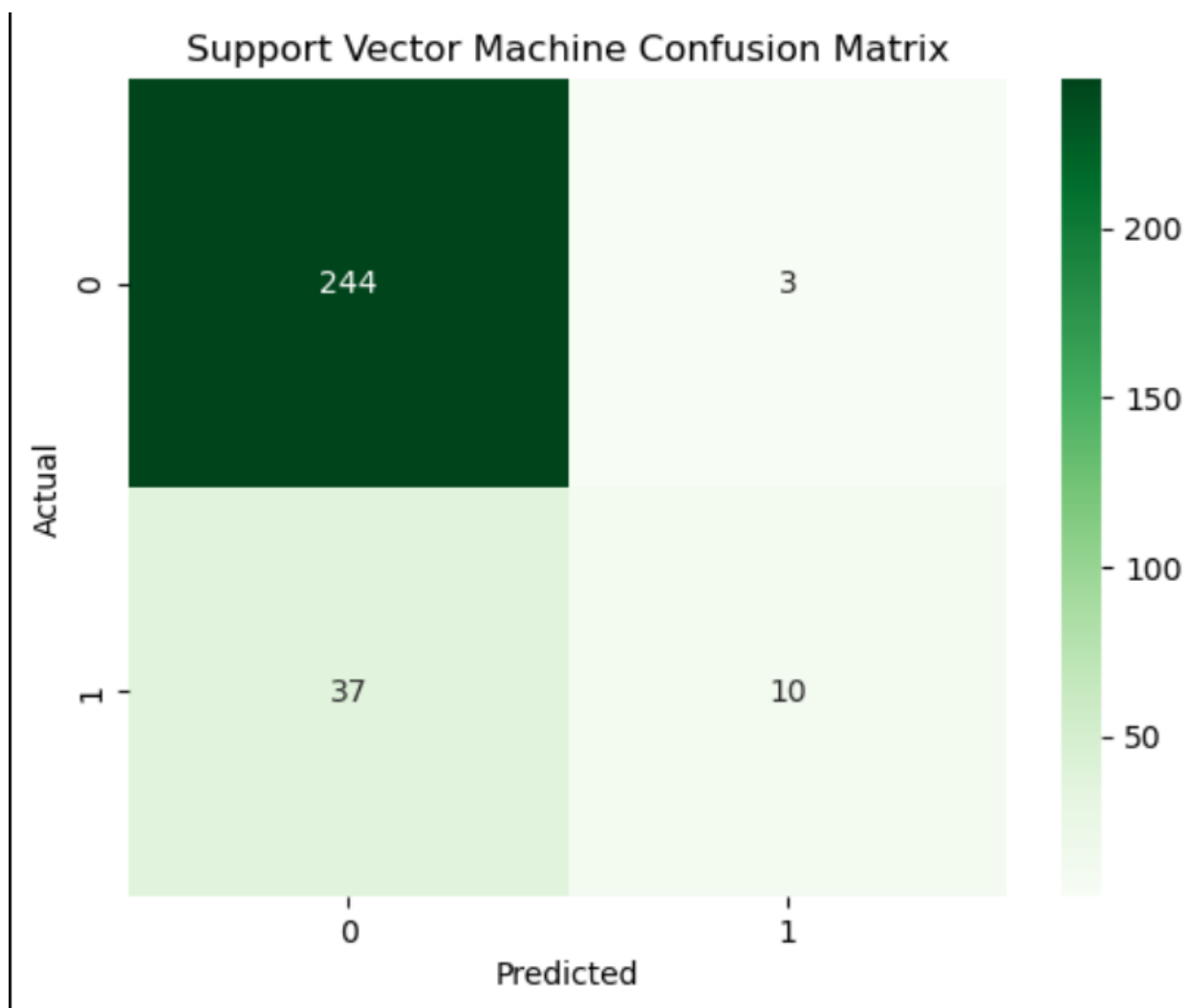
Exploratory Data Analysis

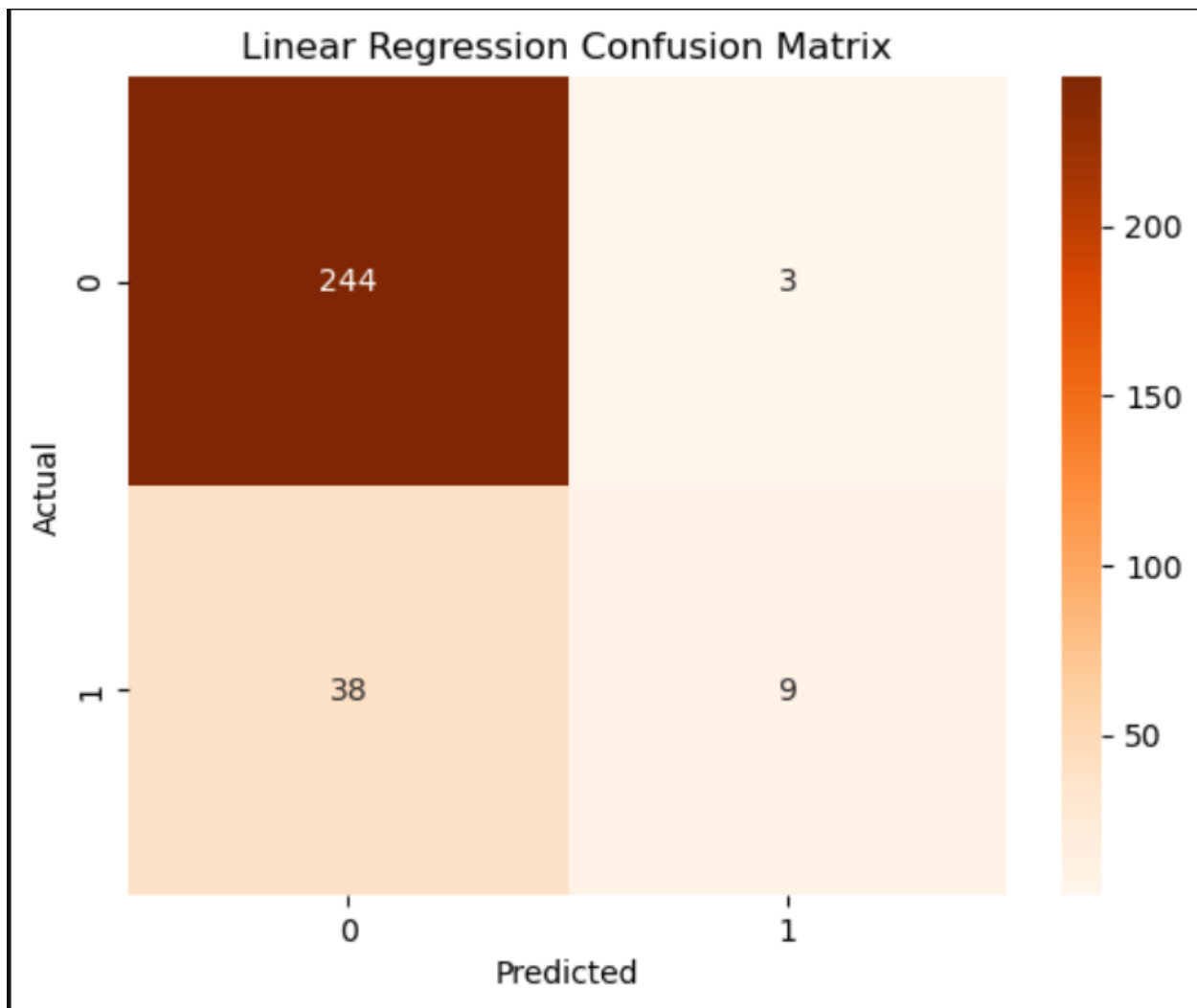
- **Trends:**
- **OverTime:** Employees working overtime suffer an attrition rate of 30.53%, compared to 10.44% for non-overtime staff.

- **Department:** Sales exhibits the highest turnover at 20.63%, followed by Human Resources at 19.05% and R&D at 13.84%.
- **Relationships:** Strong negative correlation exists between JobLevel / TotalWorkingYears and Attrition.
- **Outliers:** Extreme continuous values observed in YearsSinceLastPromotion and MonthlyIncome representing senior leadership roles.
- **Important Observations:** A significant target class imbalance exists, with 83.88% retention (No) and 16.12% attrition (Yes).

Machine Learning







Dashboard Summary

- **Dashboard Design:** Built around a dark ruby theme featuring a 3-tab executive layout: Overview, Analytical Drivers, and Predictive Risk Modeling.
- **KPIs:** Key metric tiles track Total Employee Count, Overall Attrition Rate, OverTime Attrition Rate, and Predicted High-Risk Personnel Count.
- **Visualizations:** Interactive bar charts (departmental turnover), violin/box plots (compensation distributions), scatter plots (age vs. monthly income), and feature importance bar graphs.
- **Predictive Insights:** Generates real-time flight-risk scores for active employees, flagging those with an attrition probability exceeding 0.50 for proactive retention planning.

Final Conclusions

- **Key Findings:** Analysis reveals a baseline employee attrition rate of 16.12%. Overtime work emerged as the single largest driver of voluntary turnover, boosting attrition rates to 30.53% compared to 10.44% for non-overtime personnel. Sales and Human Resources departments face the highest turnover rates at 20.63% and 19.05%, respectively.
- **Actionable Recommendations:**
 - Rebalance workloads in high-turnover roles to curb continuous overtime reliance.
 - Implement compensation reviews and flexible remote-work options for employees with long commutes.
 - Leverage the high-performing predictive model (ROC-AUC 0.806) within HR workflows to identify at-risk staff early and execute targeted stay-interviews.